prac01

Tuesday, 3 March 2026 3:02 PM

Sets

F Foods

N nutrients

Data

m_n minimum requirement for nutrient $n \in N$

c_f cost (\$/100g) for food $f \in F$

u_{fn} amount of nutrient $n \in N$ in 100g of food $f \in N$

a_n maximum requirement for nutrient $n \in N$

Variables

x_f amount (100g) of food $f \in F$

Objective

Minimize cost $= c_f \cdot x_f$ for $f \in F$ $= \sum_{f \in F} c_f \cdot x_f$

Constraints

$\sum_{f \in F} u_{fn} x_f \in [m_n, a_n] \quad \forall n \in N$ s.t. an a_n exists $\Rightarrow$ i.e. in min, max

$x_f \geq 0 \quad \forall f \in F$

Sets

F Foods

N Nutrients

Data

c_f calories (kcal) for f in Foods

p_f protein (g) for f in Foods

m_f minerals ( ) for f in Foods

v_f vitamins ( ) for f in Foods

Variables

x_i amount of food | food group $f \in F$ to make

Constraints

min_i minimum unit amount for d in Data

max_i maximum amount for d in Data

Objective

Find an optimal diet in terms of
minimal cost and meeting minimum
calorie/protein/mineral/vitamin requirements

maximize calories|protein|minerals|vitamin
while minimising cost